

Background.

Let f be analytic on $B_r(z_0) \setminus \{z_0\}$. Then, there is the Laurent's Expansion:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n \cdot (z-z_0)^n \quad (0 < |z-z_0| < r)$$

where $a_n = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{(z-z_0)^{n+1}} dz$ ($n \in \mathbb{Z}$), C is any closed contour around z_0 .

So that $a_{-1} = \frac{1}{2\pi i} \cdot \int_C f(z) dz$. Therefore, if we can find the Laurent's expansion, then we can evaluate integral.
 $\stackrel{\text{def}}{=} \text{Res}(f; z_0)$

Thm. Suppose that f is analytic inside and on a simple closed contour C , except for isolated singularities at $z_1, \dots, z_n$ inside C . Then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k).$$

There are two useful Lemmas.

Lemma. If f has a pole of order m at z_0 , then

$$\text{Res}(f, z_0) = \frac{1}{(m-1)!} \cdot \lim_{z \rightarrow z_0} \left(\frac{d}{dz} \right)^{m-1} (z-z_0)^m f(z).$$

Proof. Observe that $f(z) = \frac{a_{-m}}{(z-z_0)^m} + \dots + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \dots$ ($a_{-m} \neq 0$)

so that $(z-z_0)^m \cdot f(z) = a_{-m} + a_{-m+1}(z-z_0) + \dots + a_{-1}(z-z_0)^{m-1} + g(z) \cdot (z-z_0)^m$, where g is analytic at z_0 .

Now, differentiating $m-1$ times, and evaluating z_0 . $\square$

Lemma. Let f and g be analytic at z_0 .

If g has a simple zero at z_0 and $f(z_0) \neq 0$, then

$$\text{Res}(f/g, z_0) = \frac{f(z_0)}{g'(z_0)}$$

Proof. Since g has simple zero at z_0 , $1/g$ has a simple pole at z_0 . $\square$

Examples.

$$\int_{|z|=\pi} \frac{e^z - 1}{1 - \cos z} dz = 4\pi i.$$

Since $\cos z = \cos\left(\frac{z}{2} + \frac{z}{2}\right) = \cos^2 \frac{z}{2} - \sin^2 \frac{z}{2} = 1 - 2\sin^2 \frac{z}{2}$, so $\frac{e^z - 1}{1 - \cos z} = \frac{e^z - 1}{2\sin^2(\frac{z}{2})}$ inside $|z| = \pi$,

And, $e^z - 1$ has only one zero at $z = 0$, and $\sin^2(\frac{z}{2}) = 0 \Leftrightarrow z = 2m\pi$ ($m \in \mathbb{Z}$)

so it has only one isolated singularity inside $|z| = \pi$, at $z_0 = 0$.

Moreover, the zero of $e^z - 1$ is order of 1, and $\sin^2(z)$ has order 2,

so the given function has a simple pole at 0. Zero of

$$\text{Now, } \lim_{z \rightarrow 0} z \cdot \frac{e^z - 1}{2\sin^2(\frac{z}{2})} = 2. \text{ Therefore, } \text{Res}(f, 0) = 2.$$

Consequently, the residue theorem gives the result.

$$\int_{|z|=\pi} \frac{\sin z}{1 - \cos z} dz$$

Since $1 - \cos z = 2\sin^2(\frac{z}{2})$ and $\sin z = 2\sin(\frac{z}{2})\cos(\frac{z}{2})$,

$$\frac{\sin z}{1 - \cos z} = \frac{\cos \frac{z}{2}}{\sin \frac{z}{2}}.$$

Example.

Evaluate the integral $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$.

Construct a Curve C as piece-wise: given $\epsilon, R > 0$,

$$C_1: Z_1(t) = \epsilon \cdot e^{-it} \quad (-\pi \leq t \leq 0)$$

$$C_2: Z_2(t) = t \quad (\epsilon \leq t \leq R)$$

$$C_3: Z_3(t) = R \cdot e^{it} \quad (0 \leq t \leq \pi)$$

$$C_4: Z_4(t) = t \quad (-R \leq t \leq -\epsilon)$$

And $C = C_1 \cdot C_2 \cdot C_3 \cdot C_4$. (The composite.)

Now, the function $f(z) = \frac{e^{iz}}{z}$ is analytic inside and on C . So,

$$\int_C f(z) dz = \sum_{i=1}^4 \int_{C_i} f(z) dz$$

$$\begin{aligned} &= \int_{-\pi}^{-\epsilon} \frac{e^{i\epsilon e^{-it}}}{\epsilon e^{-it}} dt + \int_{\epsilon}^R \frac{e^{it}}{t} dt + \int_0^\pi \frac{e^{iR e^{it}}}{R e^{it}} \cdot (-i) \cdot R \cdot e^{it} dt + \int_0^\pi \frac{e^{iR \cdot e^{it}}}{R \cdot e^{it}} \cdot R \cdot i \cdot e^{it} dt \\ &= \int_{\epsilon}^R \frac{e^{it} - e^{-it}}{t} dt - i \cdot \int_{-\pi}^0 e^{iR \cdot e^{it}} dt + i \cdot \int_0^\pi e^{iR \cdot e^{it}} dt. \end{aligned}$$

$= 0$ (Cauchy's Theorem).

$$\begin{aligned} \text{Meanwhile, } \left| i \cdot \int_0^\pi e^{iR \cdot e^{it}} dt \right| &\leq \int_0^\pi |e^{iR \cdot e^{it}}| dt \leq \int_0^\pi e^{-R \sin \theta} d\theta = 2 \cdot \int_0^{\frac{\pi}{2}} e^{-R \sin \theta} d\theta \\ &\stackrel{0 \leq t \leq \frac{\pi}{2} \Rightarrow \sin t \geq \frac{2}{\pi} t}{\leq} 2 \cdot \int_0^{\frac{\pi}{2}} e^{-\frac{2R\theta}{\pi}} d\theta = 2 \cdot \left(-\frac{\pi}{2R} \right) \cdot (e^{-R} - 1) \\ &= \frac{1 - e^{-R}}{R} \cdot \pi \rightarrow 0 \text{ as } R \rightarrow \infty. \end{aligned}$$

And, $e^{ie^{it}} = \sum_{n=0}^{\infty} \frac{1}{n!} (ie^{it})^n$ follows $|e^{ie^{it}} - 1| \leq \sum_{n=1}^{\infty} \frac{1}{n!} \cdot \epsilon^n = e^\epsilon - 1$, thus

$$\int_{-\pi}^0 e^{ie^{it}} dt = \int_{-\pi}^0 e^{ie^{it}} - 1 dt + \int_{-\pi}^0 1 dt \rightarrow \pi \text{ as } \epsilon \rightarrow 0.$$

Consequently, letting $R \rightarrow \infty$ and $\epsilon \rightarrow 0$, we obtain

$$\begin{aligned} \int_0^\infty \frac{e^{ix} - e^{-ix}}{x} dx &= i\pi. \quad \text{Since } \sin x = (e^{ix} - e^{-ix})/2i, \text{ we obtain} \\ \int_0^\infty \frac{\sin x}{x} dx &= \frac{\pi}{2}. \end{aligned}$$

Thm. Suppose that f is analytic in $\mathbb{C}_\infty$, except for iso, singularities $z_1, \dots, z_n, \infty$.

Then,
$$\text{Res}(f, \infty) + \sum_{k=1}^n \text{Res}(f, z_k) = 0.$$

Proof. By the hypothesis, for some $\delta > 0$, a Laurent's representation

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

is valid in $\delta < |z| < \infty$. Specifically, put $\delta = \max\{|z_k| \mid k=1, 2, \dots, n\}$.

Choose $R > \delta$, and γ be a curve, $\{z \in \mathbb{C} \mid |z| = R\}$, the clockwise direction.

And, for $|z| = R$, the Laurent's representation of f converges uniformly, so that

$$\int_{\gamma} f(z) dz = \sum_{n=-\infty}^{\infty} a_n \int_{\gamma} z^n dz = -2\pi i \cdot a_{-1}.$$

so, define $\text{Res}(f, \infty) = -a_{-1}$.

Moreover, $f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$ is analytic for $|z| > R$

iff $f(\frac{1}{z}) = \sum_{n=-\infty}^{\infty} a_n z^{-n}$ is analytic for $0 < |z| < \frac{1}{R}$ ($R < \frac{1}{|z|} \Leftrightarrow |z| < \frac{1}{R}$).

So $a_{-1} = \text{Res}\left(\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right), 0\right)$. $\downarrow$ Actually, just definition.

Example. Let $f(z) = 1/(z-3)(z^n-1)$ ($n \in \mathbb{N}$). Evaluate $I = \frac{1}{2\pi i} \int_{|z|=2} f(z) dz$.

By the Residue Theorem, $I = \sum_{i=1}^n \text{Res}(f, \exp(2\pi i \cdot \frac{k}{n})) = -\text{Res}(f, \infty) - \text{Res}(f, 3)$

Since $\text{Res}(f, \infty) = -\text{Res}\left(\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right), 0\right) = -\text{Res}\left(\frac{1}{z^2} \cdot \frac{z^{n+1}}{(1-z)(1-z^n)}, 0\right) = 0,$

because the rational function is analytic at $z=0$.

Consequently,

$$I = -\text{Res}(f, 3) = -\lim_{z \rightarrow 3} \frac{1}{z^{n-1}} = \frac{1}{1-3^n},$$

because f has simple pole at $z=3$.

